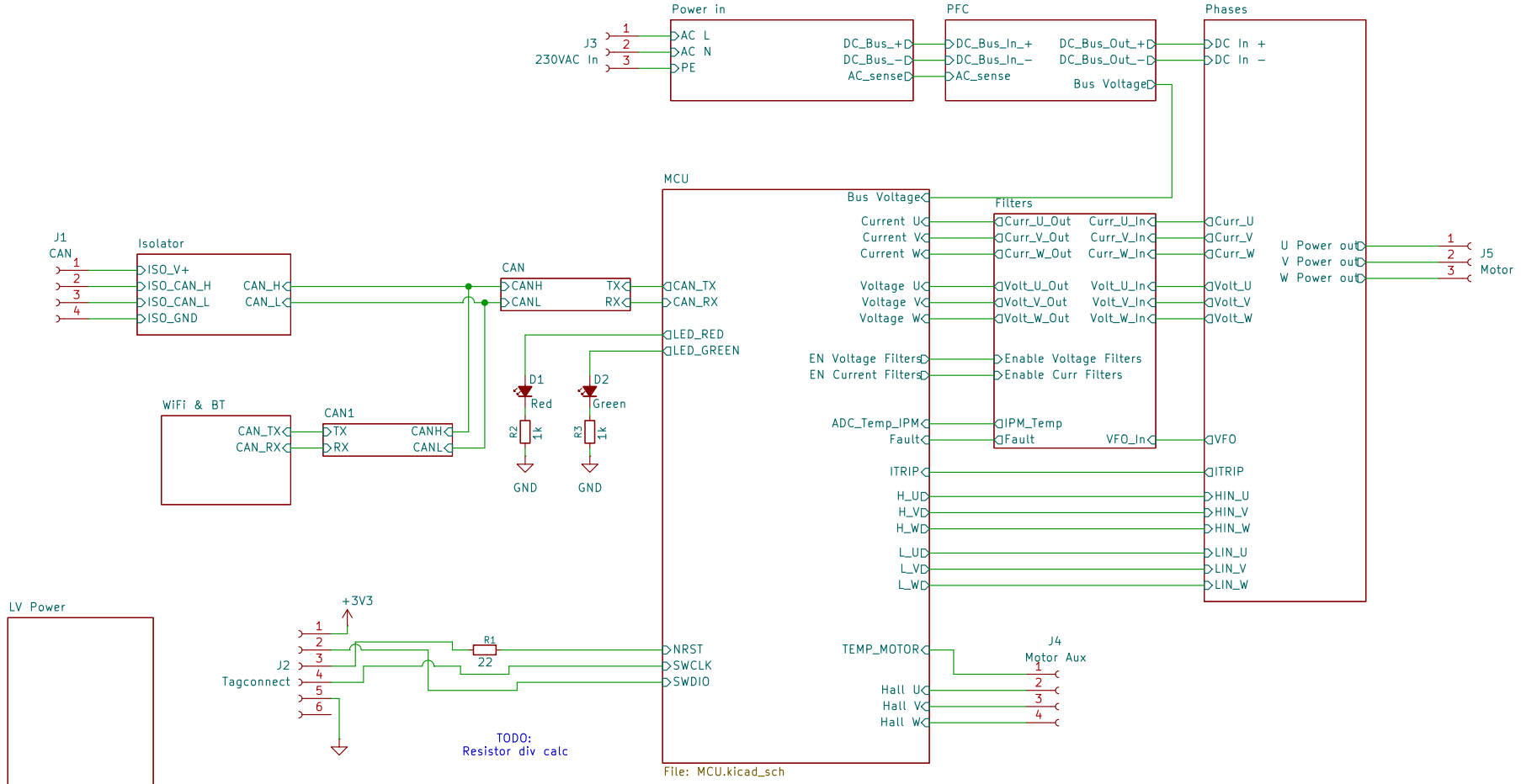



# HVAC VESC

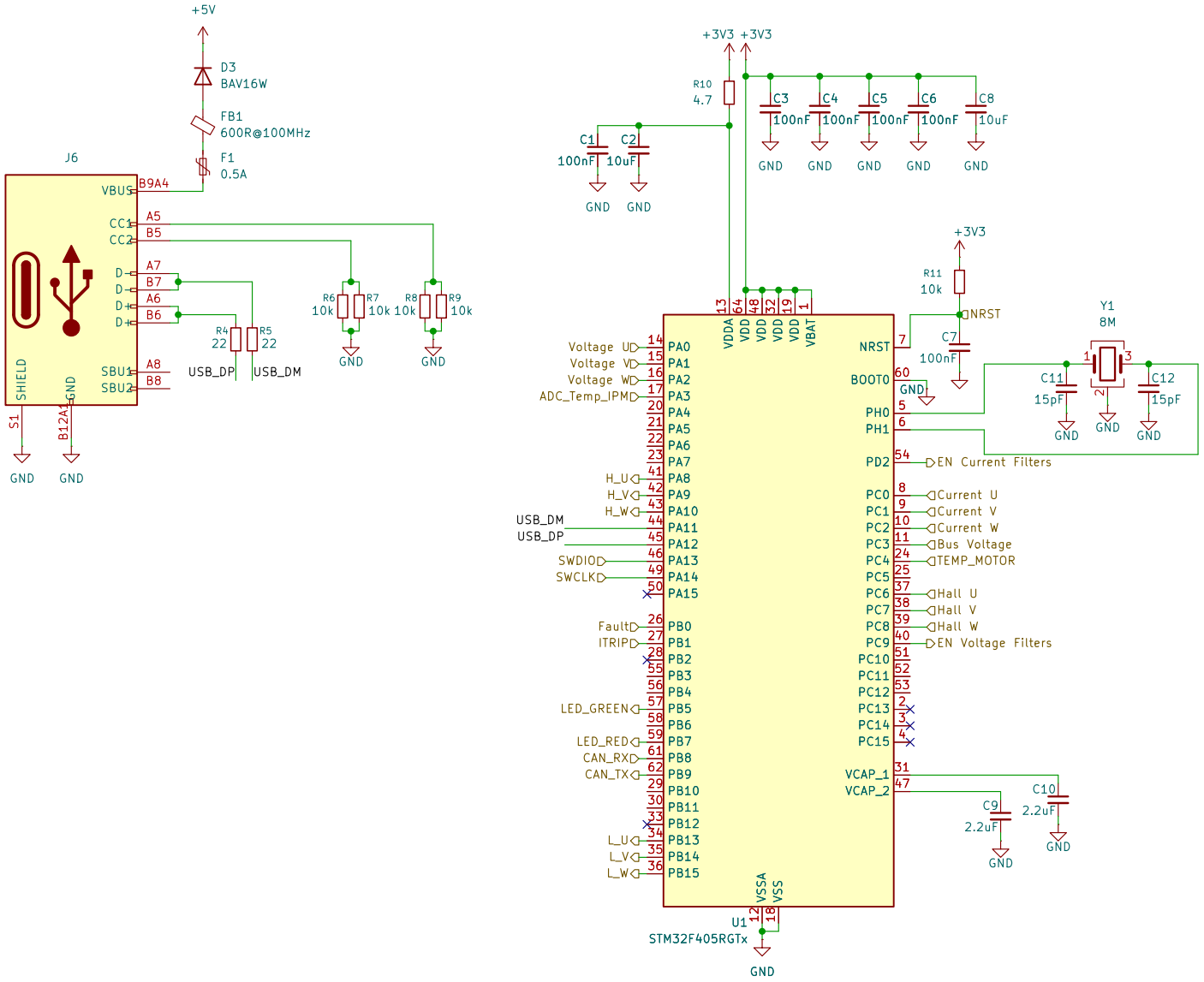
Needs to follow for CE cert:
EN IEC 63000 (RoHS)
IEC 60335-2-34 (Particular requirements for motor-compressors)
IEC_61000-3-2 (Limits for harmonic current emissions) <--- Most important

**Picobello B.V.**

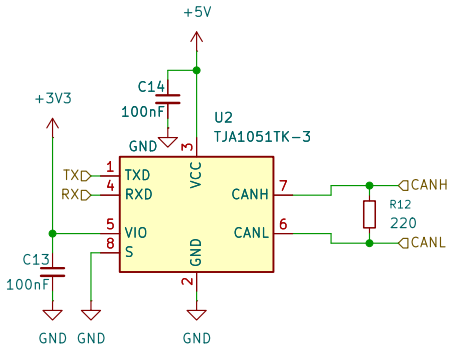
Sheet: /
File: CompressorController.kicad_sch

***Title: Mains Powered VESC MD***

| Size: A4 | Date: 2026-04-16 | **Rev: R1** |
|---|---|---|
| KiCad E.D.A. 10.0.0 | | Id: 1/11 |




| | | |
|---|---|---|
| **Picobello B.V.** | | |
| Sheet: /MCU/ | | |
| File: MCU.kicad_sch | | |
| ***Title: Mains Powered VESC MD*** | | |
| Size: A4 | Date: 2026-04-16 | **Rev: R1** |
| KiCad E.D.A. 10.0.0 | | Id: 2/11 |




| | | |
|---|---|---|
| **Picobello B.V.** | | |
| Sheet: /CAN1/ | | |
| File: can.kicad_sch | | |
| ***Title: Mains Powered VESC MD*** | | |
| Size: A4 | Date: 2026-04-16 | **Rev: R1** |
| KiCad E.D.A. 10.0.0 | | Id: 3/11 |

DC In + — DC Bus +

DC In − — DC Bus −

+15V

C18 2.2uF — GND
C19 100nF — GND
C23 1nF — GND

U6
IKCM10H60GAXKMA1

VDD 13
VSS 16 — GND
NC 24 — GND
VS(U) 1 — C20 10uF / C24 100nF
VB(U) 2
VS(V) 3 — C21 10uF / C25 100nF
VB(V) 4
VS(W) 5 — C22 10uF / C26 100nF
VB(W) 6
HIN(U) 7 — HIN_UD — R13 ZZ
HIN(V) 8 — HIN_VD — R14 ZZ
HIN(W) 9 — HIN_WD — R15 ZZ
LIN(U) 10 — LIN_UD — R16 ZZ
LIN(V) 11 — LIN_VD — R17 ZZ
LIN(W) 12 — LIN_WD — R18 ZZ
VFO 14 — R19 10k / R20 10k / R21 10k / R22 10k / R23 10k / R24 10k — GND
ITRIP 15 — VFO — 1k R25 — C27 1nF — GND
P 23 — DC Bus +
U 22
V 21
W 20
NU 19
NV 18
NW 17 — DC Bus −

C28 470nF

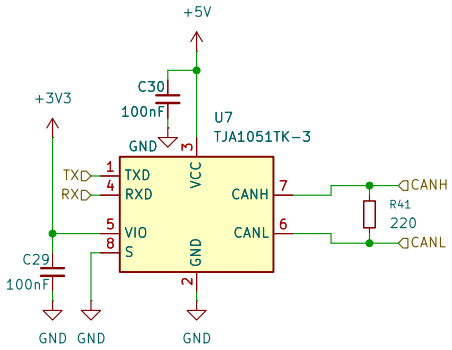
IKCM10H60GA
C3615235

U Power out — R29 200k / R30 200k / R31 200k / R32 3.16k — GND — Volt_U
V Power out — R33 200k / R34 200k / R35 200k / R36 3.16k — GND — Volt_V
W Power out — R37 200k / R38 200k / R39 200k / R40 3.16k — GND — Volt_W

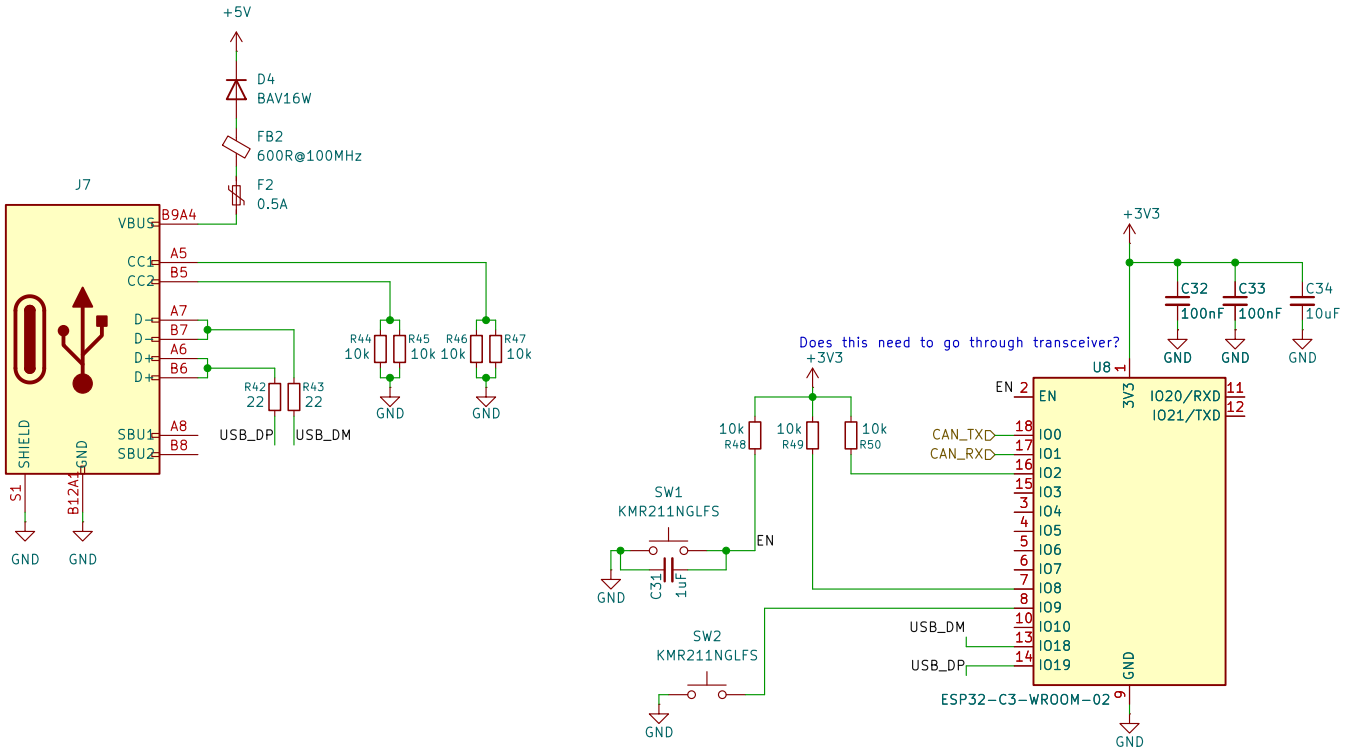
TODO Resistor Div calc

+3V3 — C15 100nF — GND
TP7 Curr_U — U3 INA240A1D — GND
R26 50mR
+3V3 — C16 100nF — GND
TP8 Curr_V — U4 INA240A1D — GND
R27 50mR
+3V3 — C17 100nF — GND
TP9 Curr_W — U5 INA240A1D — GND
R28 50mR

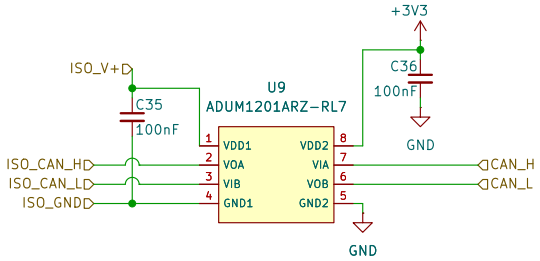
Picobello B.V.
Sheet: /Phases/
File: phases.kicad_sch
Title: Mains Powered VESC MD
Size: A3 | Date: 2026-04-16 | Rev: R1
KiCad E.D.A. 10.0.0 | Id: 4/11




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|---|---|---|
| **Picobello B.V.** | | |
| Sheet: /CAN/ | | |
| File: can.kicad_sch | | |
| ***Title: Mains Powered VESC MD*** | | |
| Size: A4 | Date: 2026-04-16 | **Rev: R1** |
| KiCad E.D.A. 10.0.0 | | Id: 5/11 |

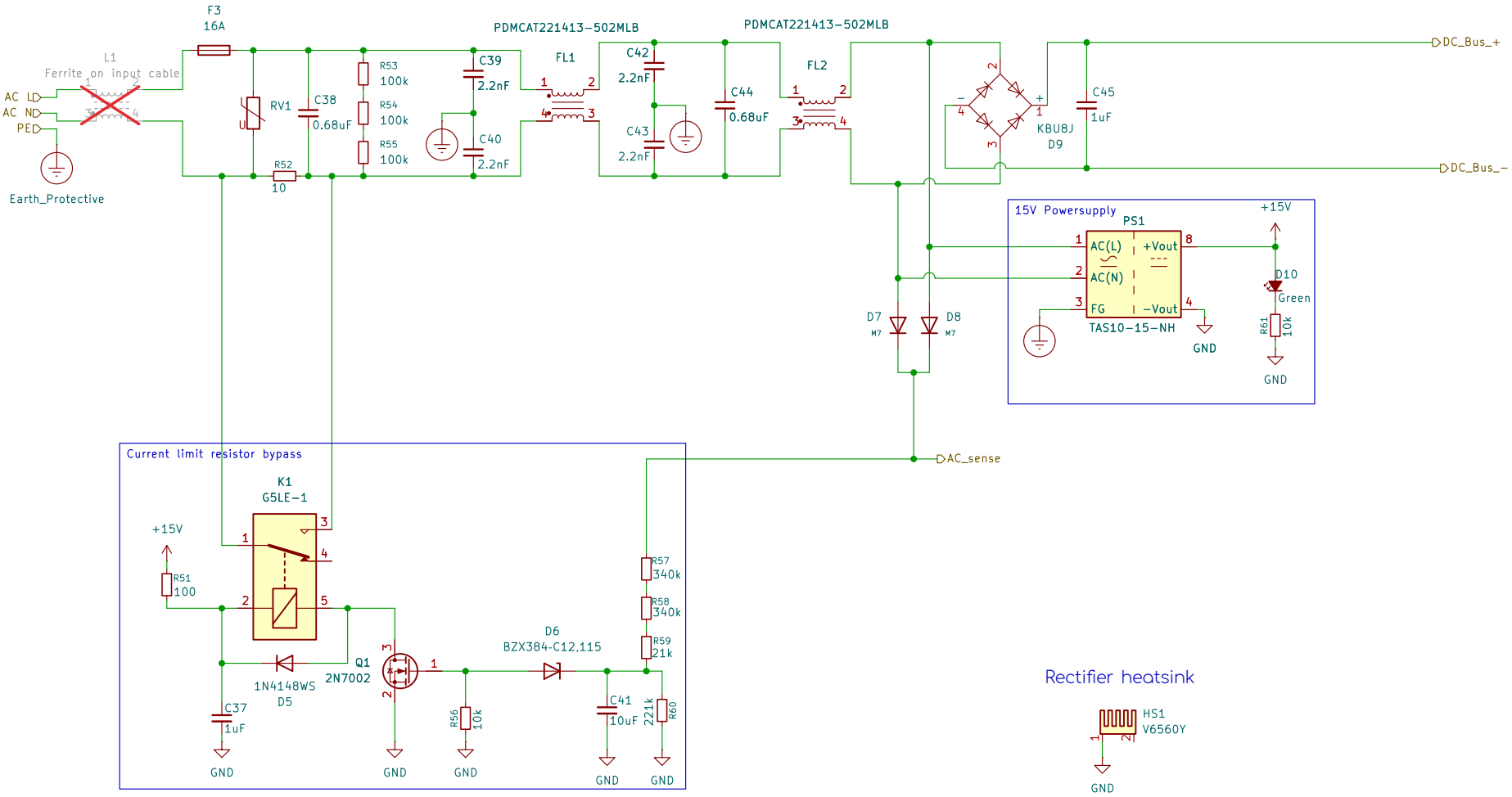



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|---|---|---|
| **Picobello B.V.** | | |
| Sheet: /WiFi & BT/ | | |
| File: express.kicad_sch | | |
| ***Title: Mains Powered VESC MD*** | | |
| Size: A4 | Date: 2026-04-16 | **Rev: R1** |
| KiCad E.D.A. 10.0.0 | | Id: 6/11 |

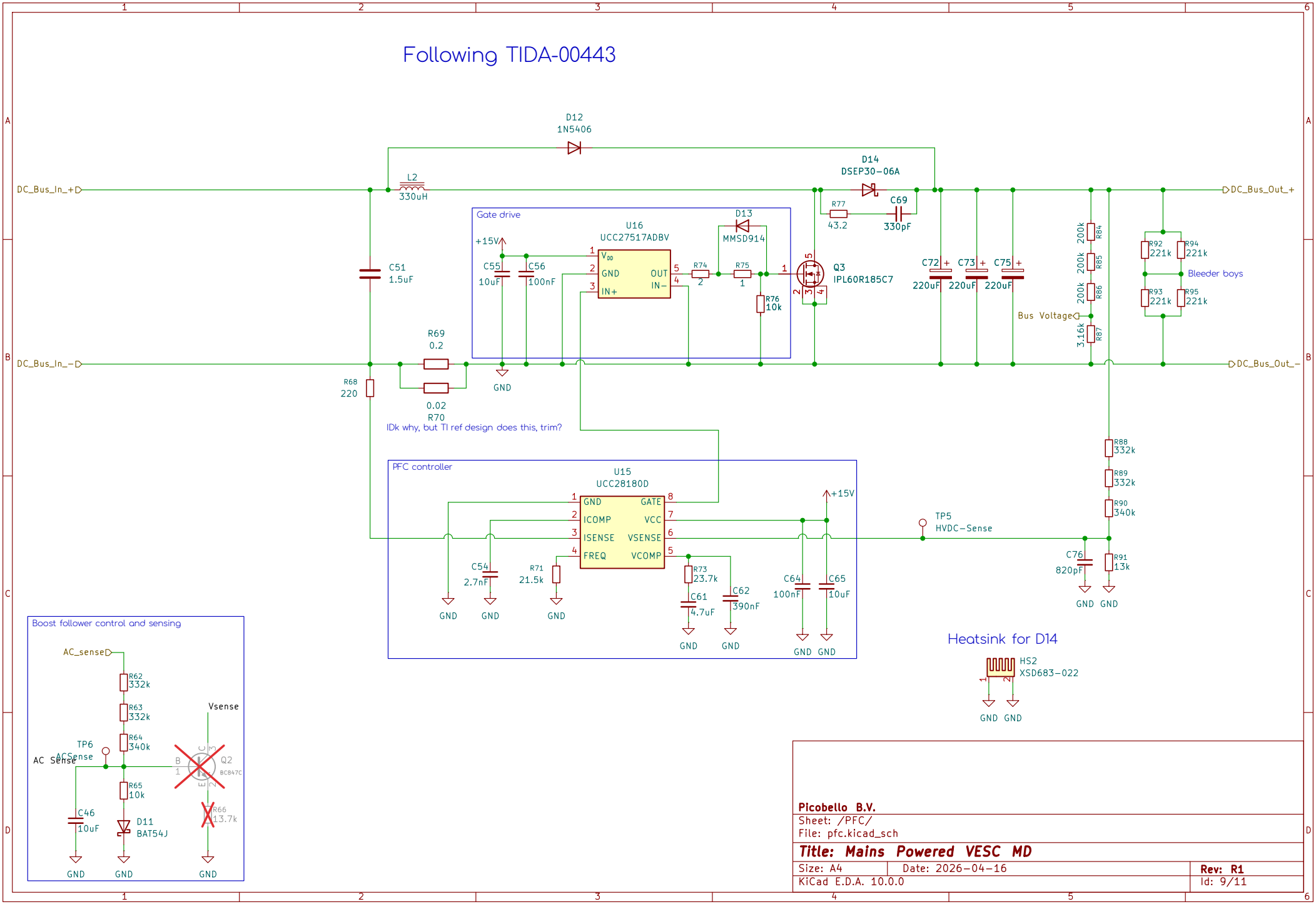


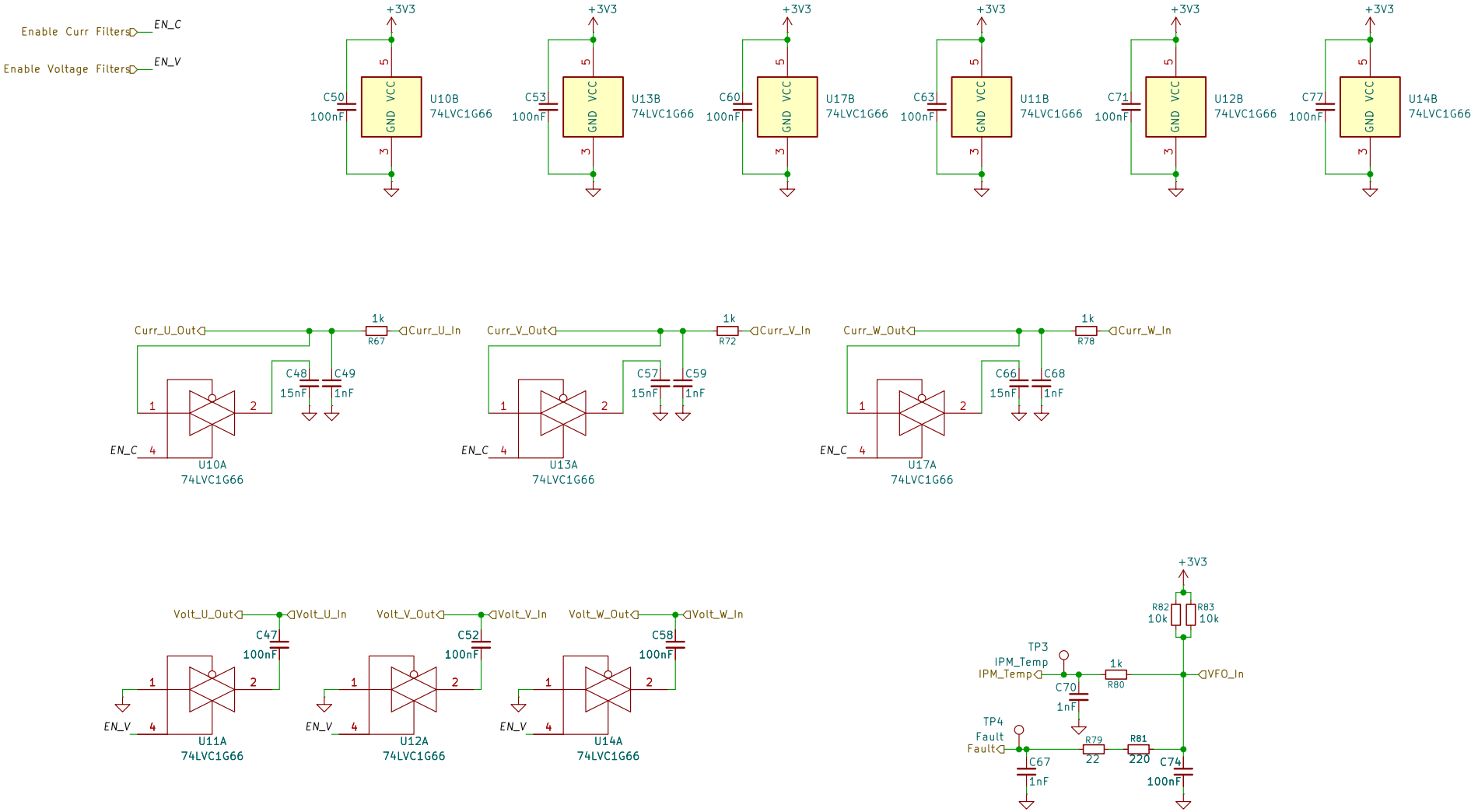

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|---|---|---|
| **Picobello B.V.** | | |
| Sheet: /Isolator/ | | |
| File: isolator.kicad_sch | | |
| ***Title: Mains Powered VESC MD*** | | |
| Size: A4 | Date: 2026-04-16 | **Rev: R1** |
| KiCad E.D.A. 10.0.0 | | Id: 7/11 |

# TIDA-00707 Reference design



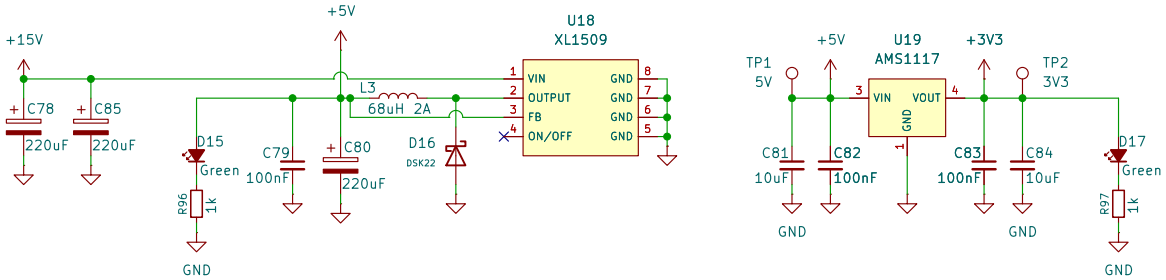

| | | |
|---|---|---|
| **Picobello B.V.** | | |
| Sheet: /Power in/ | | |
| File: powerin.kicad_sch | | |
| ***Title: Mains Powered VESC MD*** | | |
| Size: A4 | Date: 2026-04-16 | **Rev: R1** |
| KiCad E.D.A. 10.0.0 | | Id: 8/11 |


Following TIDA-00443
D12
1N5406
D14
DSEP30-06A
L2
330uH
DC_Bus_In_+
DC_Bus_In_-
DC_Bus_Out_+
DC_Bus_Out_-
R77
43.2
C69
330pF
Gate drive
U16
UCC27517ADBV
+15V
C55
10uF
C56
100nF
D13
MMSD914
R74
2
R75
1
R76
10k
Q3
IPL60R185C7
C72
220uF
C73
220uF
C75
220uF
C51
1.5uF
R69
0.2
R70
0.02
R68
220
GND
IDk why, but TI ref design does this, trim?
R84
200k
R85
200k
R86
200k
R87
3.16k
Bus Voltage
R92
221k
R94
221k
R93
221k
R95
221k
Bleeder boys
PFC controller
U15
UCC28180D
GND
ICOMP
ISENSE
FREQ
GATE
VCC
VSENSE
VCOMP
C54
2.7nF
R71
21.5k
R73
23.7k
C61
4.7uF
C62
390nF
C64
100nF
C65
10uF
TP5
HVDC-Sense
R88
332k
R89
332k
R90
340k
R91
13k
C76
820pF
Heatsink for D14
HS2
XSD683-022
Boost follower control and sensing
AC_sense
R62
332k
R63
332k
R64
340k
R65
10k
TP6
ACSense
AC Sense
Q2
BC847C
Vsense
R66
13.7k
C46
10uF
D11
BAT54J
Picobello B.V.
Sheet: /PFC/
File: pfc.kicad_sch
Title: Mains Powered VESC MD
Size: A4
Date: 2026-04-16
KiCad E.D.A. 10.0.0
Rev: R1
Id: 9/11




VFO Pin is also UVLO and OC fault output

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|---|---|
| **Picobello B.V.** | |
| Sheet: /Filters/ | |
| File: filters.kicad_sch | |
| ***Title: Mains Powered VESC MD*** | |
| Size: A4 | Date: 2026-04-16 |
| KiCad E.D.A. 10.0.0 | **Rev: R1** |
| | Id: 9/11 |







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|---|---|---|
| **Picobello B.V.** | | |
| Sheet: /LV Power/ | | |
| File: lvpower.kicad_sch | | |
| ***Title: Mains Powered VESC MD*** | | |
| Size: A4 | Date: 2026-04-16 | **Rev: R1** |
| KiCad E.D.A. 10.0.0 | | Id: 10/11 |